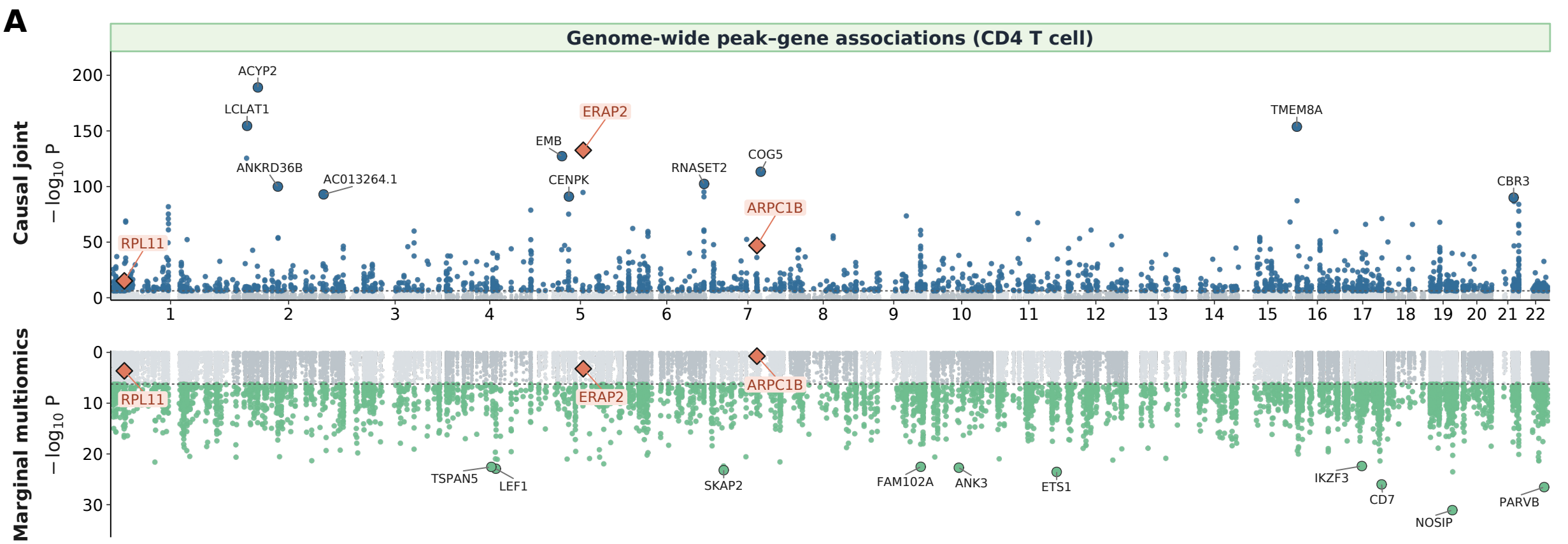
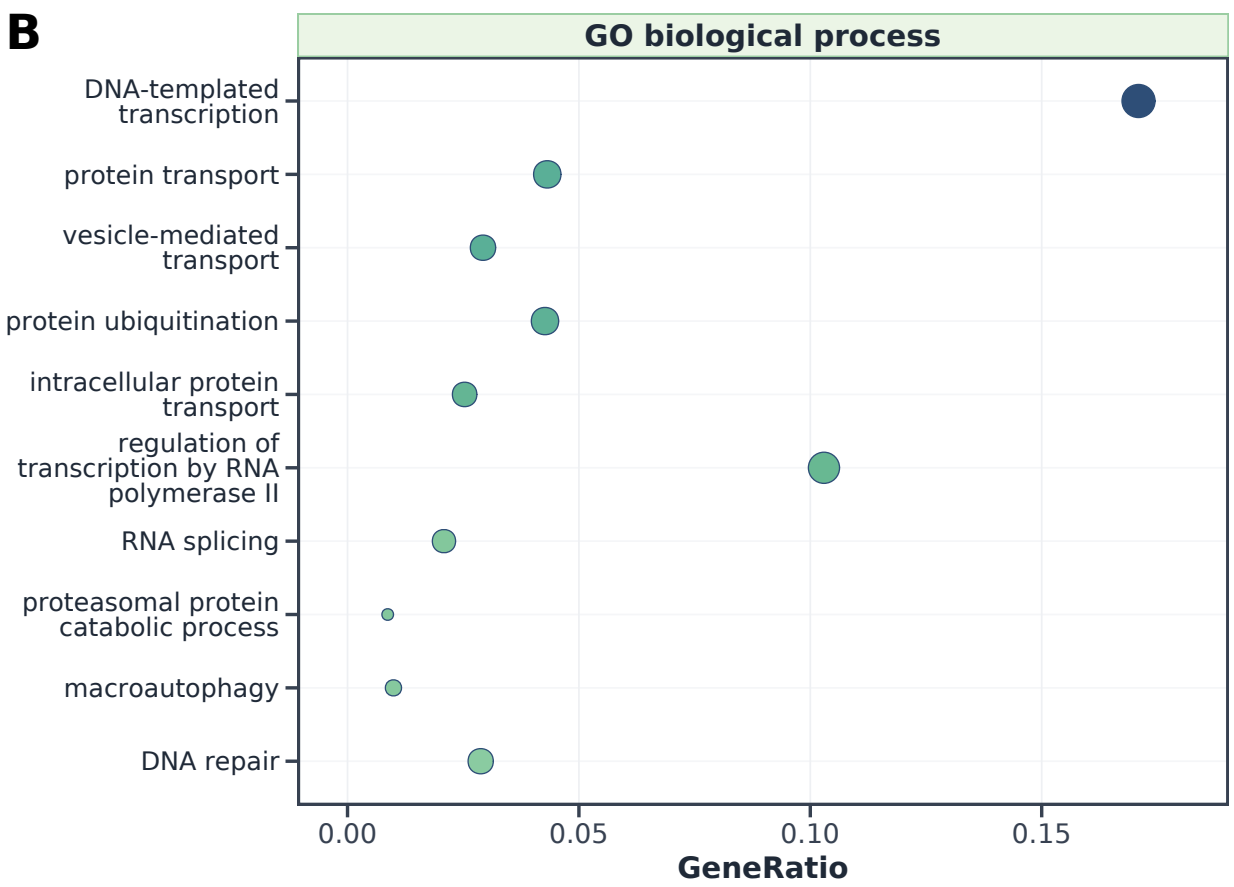
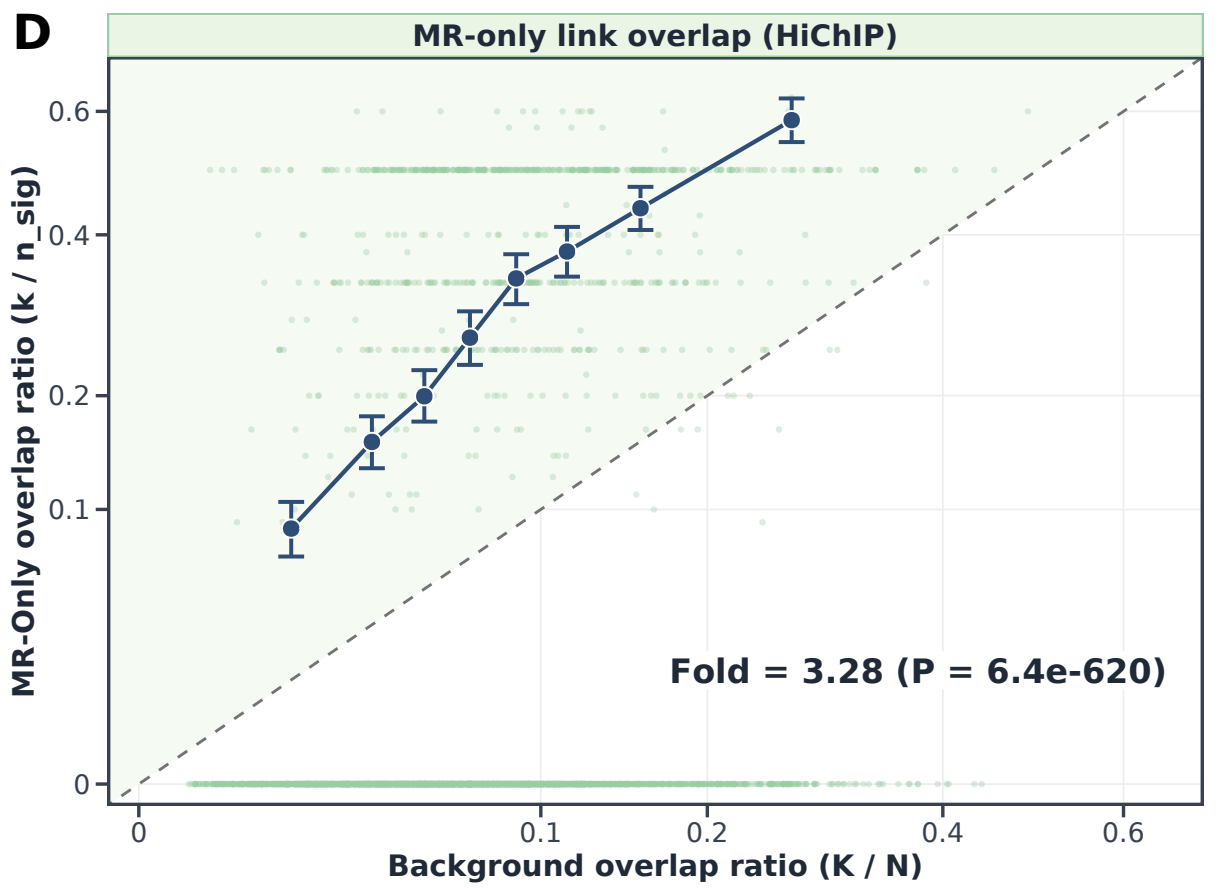
A


Genome-wide peak-gene associations (CD4 T cell)
Causal joint
− log10 P
ACYP2
LCLAT1
ANKRD36B
AC013264.1
ERAP2
EMB
CENPK
RNASET2
COG5
ARPC1B
TMEM8A
CBR3
RPL11
Marginal multiomics
TSPAN5
LEF1
SKAP2
FAM102A
ANK3
ETS1
IKZF3
CD7
NOSIP
PARVB

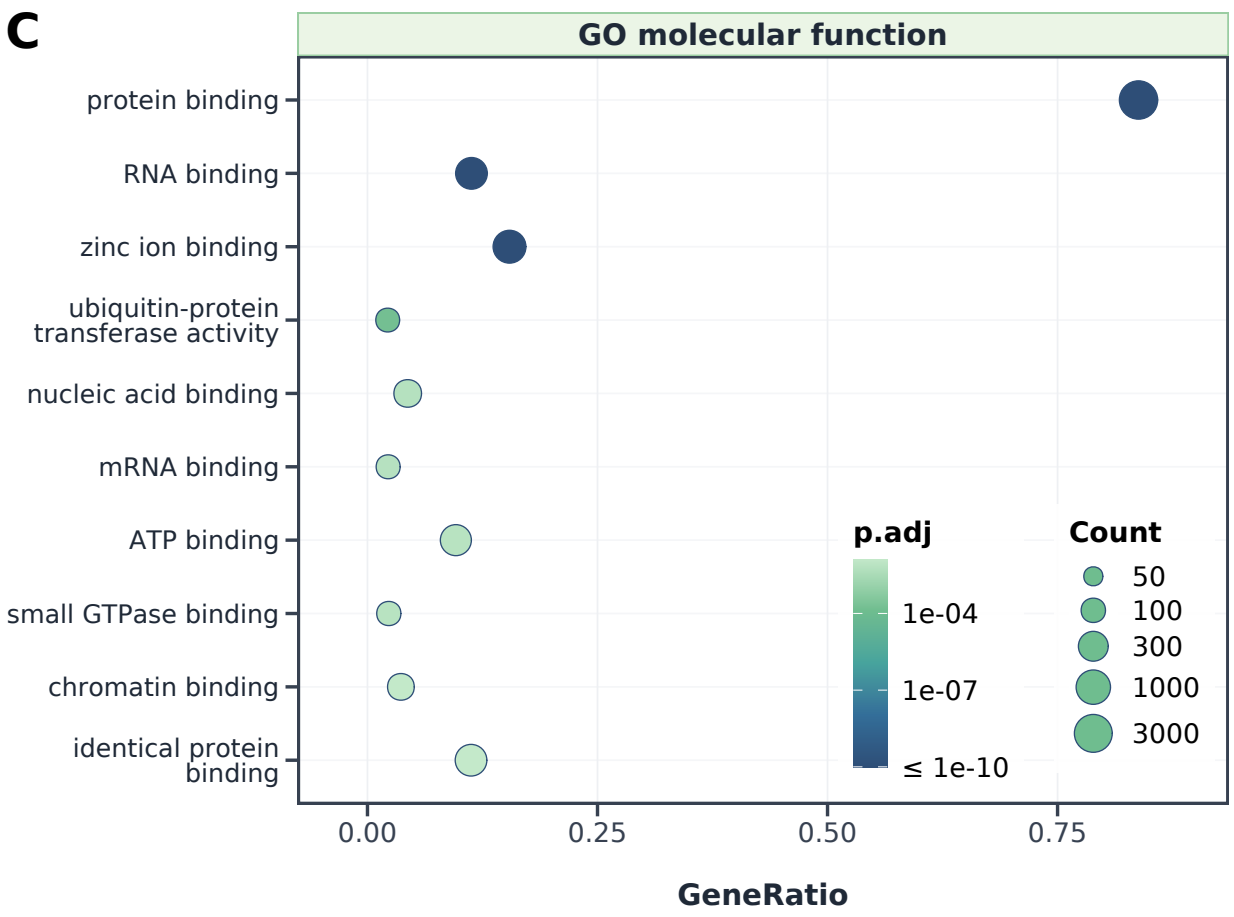
B


GO biological process
DNA-templated transcription
protein transport
vesicle-mediated transport
protein ubiquitination
intracellular protein transport
regulation of transcription by RNA polymerase II
RNA splicing
proteasomal protein catabolic process
macroautophagy
DNA repair
GeneRatio

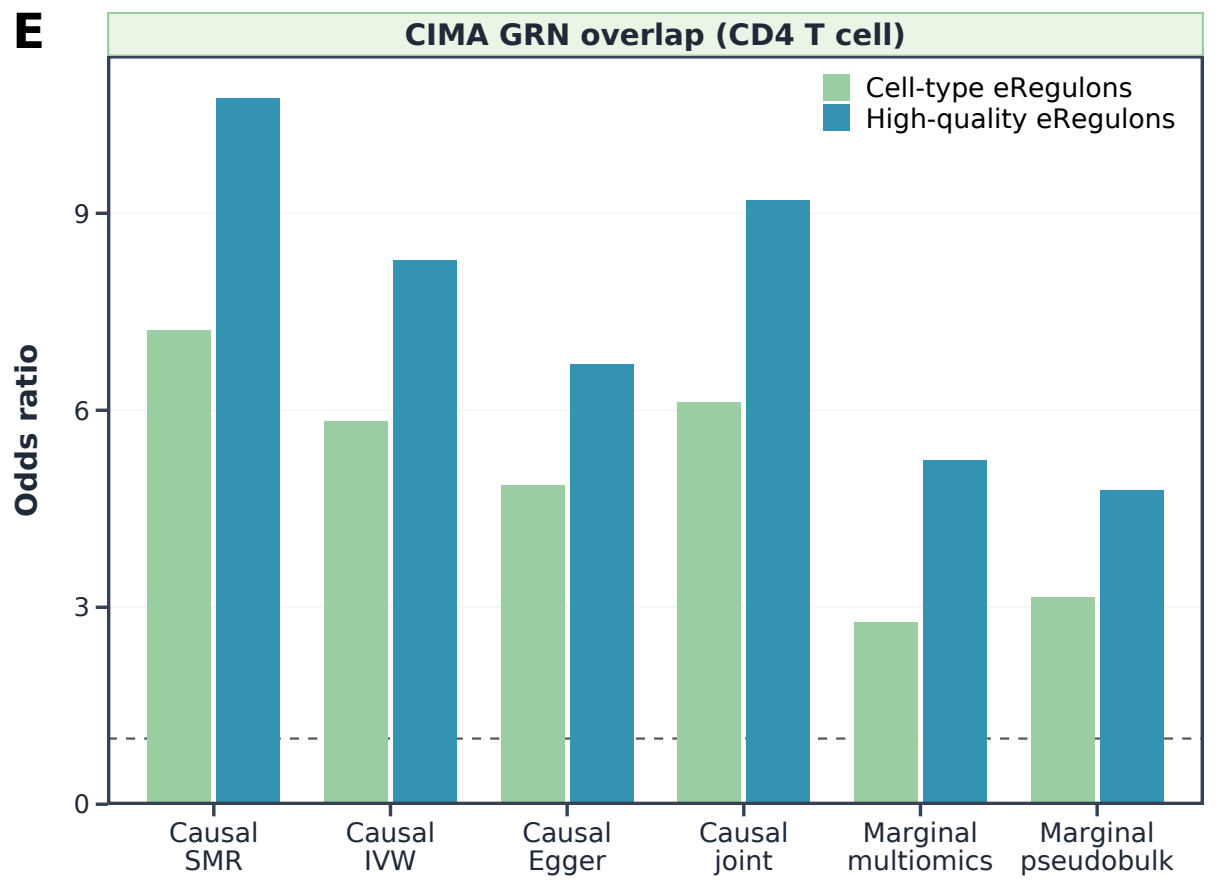
D


MR-only link overlap (HiChIP)
MR-Only overlap ratio (k / n_sig)
Fold = 3.28 (P = 6.4e-620)
Background overlap ratio (K / N)

C


GO molecular function
protein binding
RNA binding
zinc ion binding
ubiquitin-protein transferase activity
nucleic acid binding
mRNA binding
ATP binding
small GTPase binding
chromatin binding
identical protein binding
p.adj
1e-04
1e-07
≤ 1e-10
Count
50
100
300
1000
3000
GeneRatio

E


CIMA GRN overlap (CD4 T cell)
Cell-type eRegulons
High-quality eRegulons
Odds ratio
Causal SMR
Causal IVW
Causal Egger
Causal joint
Marginal multiomics
Marginal pseudobulk